BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

MISS THANANYA OSOCHPROMMA
MR. SUTTIKARN KHUNTONG
MR.PATHANIN OPACHALEARN

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
MAHIDOL UNIVERSITY
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Computer Engineering Project

entitled

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Thesis
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BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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on
April 5, 2026

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บทคัดย่อ

บทคัดย่อภาษาไทย (not required for your project proposal, but it is mandatory for the black book)

คำสำคัญ : MLP

29 หน้า

BUDDY: AN INTERACTIVE COMPANION ROBOT FOR FAMILY

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DATE OF GRADUATION April 5, 2026

ABSTRACT

An abstract (250-400 words) should provide a concise summary of your entire thesis. It should report significant elements of your thesis including background or introduction in brief, objectives, statistical data, key findings, and conclusion. Mathematical formulas, diagrams, and other illustrativematerials are not recommended for inclusion. A strong abstract should be self-contained; without abbreviations, footnotes, references. Outside readers typically view the abstract before deciding to read the thesis, so it should be well written, logical, and a complete reflection on you work. The abstract is typically written last after finishing chapter 4 and 5.

KEYWORDS : MLP

29 Pages

CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

1.1 Background

Background Loneliness and social isolation are rising, particularly among the elderly population. It is concerning as Thailand is entering an aged society, a community in which the percentage of people 60 and older makes up 20% or more of the total population in a certain area, or when the percentage of people 65 and older makes up 14% or more of the total population in the same location [1]. According to the National Statistical Office Ministry of Digital Economy and Society, as of 2024 (the latest update at the moment), the elderly population has risen to 20% [2]. There are ways to take care of elders when the main caretaker can not be there as much as they want, like taking them to a nursing home or hiring a caretaker. However, most of the elders prefer to stay home, and it is hard to find people who are trustworthy for your loved one. Elder people might not be enthusiastic about using a smartphone, it is even harder when their caretakers have to go outside, and there is no way to communicate with elders. Implementing IoT in the household might be a choice, for instance, you can buy a security camera to monitor them. Most security cameras indeed allow you to do two-way communication, but it is not ideal, as it contains a lot of feedback. Moreover, to allow them to also communicate with you, you have to implement more things. According to the provided information, we think that it would be better if the elders could be in their own homes with a companion that helps monitor them, and let them connect with their caretakers by voice or an easy-to-access button.

1.2 Objective

1. To introduce a video call function .
2. To develop a emotion progress report

3. To enhance Human-Robot Interaction (HRI)
4. To develop elder friendly hardware

1.3 Scope

1. Integrating GEMMA 3 LLM for more natural and engaging dialogues with the elderly.
2. Establishing an interactive system using a tablet's application for data retrieval and feedback
3. Building an intercommunication system using ZEGOCLOUD.
4. Analyzing elder's mood progress report , but it won't diagnose mental health

1.4 Expected Results

1. To develop a friendly companion designed to engage with elders.
2. To enable video call function for real-time communication.
3. To provide insights into the elder's mood progress report for the caretaker.

1.5 Timeline

Table 1.1 Project Timeline

Plan	**Timeline**								
	2025								
	Apr	**May**	**Jun**	**Jul**	**Aug**	**Sep**	**Oct**	**Nov**	**Dec**
1: Finish with project proposal									
2: Design the application									
3: Testing									
4: Fixing the problem									
5: Continue on to develop									
6: Submit the									

CHAPTER 2

LITERATURE REVIEW

Technology has become an essential tool in addressing the challenges associated with an aging population. One area of growing interest is the development of smart companion systems aimed at supporting elderly individuals in their daily lives. These systems integrate artificial intelligence and communication platforms to enhance quality of life, improve safety, and foster independence. This research aims to provide personalized and accessible care for elderly individuals and their families, supporting their emotional well-being. This literature review investigates the application of each selected study to the development of a smart companion system for elderly care. The primary goals of the study are to implement a function to predict emotions based on audio and facial recognition data, conduct natural conversations, introduce a video call function for real-time communication, and develop a monitoring system that functions as a camera for elderly care. These features aim to meet both the emotional and practical needs of elderly individuals, offering comprehensive support. The scope of this research is shaped by the areas explored in the selected studies. The Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach focuses on emotion recognition using multimodal inputs by converting audio and visual features into structured text, offering a new approach for LLM-based classification accuracy. The AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models investigates emotionally-aware dialogue systems through deep learning-based sentiment analysis and dialogue generation. The Development of an Intelligent Dialogue Agent for Older Adults explores how reinforcement learning techniques can adapt conversation topics based on user preferences to promote social companionship. These studies collectively provide insights into multimodal emotion recognition, sentiment prediction, and personalized dialogue generation, forming the foundation for this research.

2.1 Multimodal Emotion Recognition Using Feature Fusion: An LLM-Based Approach [2]

Multimodal emotion recognition has evolved from traditional machine learning methods to deep learning-based multimodal fusion techniques, integrating speech, facial expressions, and textual data. While deep learning models like CNNs, RNNs, and transformers have improved classification accuracy, they often suffer from high computational costs and feature alignment challenges. Recent advancements in Large Language Models (LLMs) such as BERT, RoBERTa, and GPT-3 have enabled emotion recognition through textual representations of multimodal inputs. This paper addresses key gaps by proposing a novel feature fusion approach that converts audio and visual features into structured text, processed by a fine-tuned DistilRoBERTa model for emotion classification. The method achieves state-of-the-art accuracy on benchmark datasets, demonstrating that LLM-based textual fusion can match or exceed traditional multimodal deep learning models while improving scalability and interpretability. This approach bridges the gap between multimodal emotion recognition and NLP, offering a promising direction for future affective computing applications.

2.2 AI Affective Conversational Robot with Hybrid Generative-Based and Retrieval-Based Dialogue Models [3]

ChatBot technology has become a widely used in various application fields. An important topic in the research on conversational robots is the improvement of their temperature during operation for enhanced user interaction. In this study, we propose an artificial intelligence affective conversational robot (AIACR), which is an integration of an artificial intelligence deep learning sentiment analysis model and generative-and retrieval-based dialogue models. The sentiment analysis model developed in this study uses three models, namely, multilayer perceptron (MLP), long short-term memory (LSTM) and bidirectional long short-term memory (BiLSTM). Moreover, word2vec and semantics are utilized as the basis for similarity ranking models. The deep learning dialogue model, sentiment analysis model, and similarity model were integrated and compared as well. The experimental results show that the sentiment analysis model, similarity model, and dialogue model respectively utilize BiLSTM, word2vec, and the retrieval-based model to achieve the best dialogue performance. The major research contributions of this study are the developed AIACR and the proposed affective conversational robot index (ACR Index) as a criterion for evaluating the effectiveness of emotional dialogue robots.

The main objectives of this paper are as follows:

- To compare the effects of conversational robots in retrieval- and generative-based models.
- To use artificial intelligence (AI) technology such as multilayer perceptron (MLP), long short-term memory (LSTM), and bidirectional LSTM (BiLSTM) to conduct sentiment analysis comparison.
- To integrate the dialogue model with the sentiment prediction model to produce a dialogue with emotions.

2.3 Development of an Intelligent Dialogue Agent for Older Adults : Evaluation of Functions to Control Spontaneous Talk and Coordinate Speech Content [5]

A Reinforcement Learning-Based Approach Intelligent Dialogue Agents (IDAs) have been increasingly utilized to support elderly care by offering social companionship and promoting preventive care activities. Traditional voice-interaction devices like Google Home and Amazon Echo provide information in response to user queries but cannot initiate spontaneous conversations. Recent research has focused on developing IDAs with spontaneous talking functions (STF) to foster more natural and friendly interactions, enhancing users' sense of familiarity and engagement. Kitakoshi et al. (2019) proposed an IDA that combines STF with a Speech-Content Coordinating Function (SCCF) to adapt speech content based on user preferences and behavioral trends. The SCCF employs reinforcement learning (RL) algorithms, specifically a policy gradient method, to select appropriate speech topics at the right moments. Experiments conducted with students demonstrated that users responded positively to personalized content but highlighted the need to optimize speech length and RL parameters to improve learning performance. The study suggests that integrating adaptive dialogue generation with preventive care systems can encourage habitual use among older adults, contributing to comprehensive preventive care frameworks.

CHAPTER 3

METHODOLOGY

This section describes the methodology for developing the smart companion system for elderly care. The methodology comprises system design, data collection, pre-processing techniques, machine learning methods, hyperparameter configurations, and performance evaluation. The technical methods used in this study include speech recognition, natural language processing, sentiment analysis, and real-time video intercommunication. Each stage of the methodology is explained to ensure clarity and reproducibility.

3.1 Caretaker Application

3.1.1 UI/UX Design

As the primary user is the caretaker who needs quick access to important information, therefore this application was designed with a focus on simplicity, and usability. The design process began with wireframing with Figma, where the layout, visual hierarchy, and user flow were all maintained by outlining each page.

An emotion analysis dashboard was created to provide caregivers with an easy to understand overview of the care receiver's emotion throughout the day, a recent mood card, and chat log between care receiver and the Buddy (care receiver application). The visualization includes graphs and datacard of the emotion trends calculated by an equation implemented inside the care receiver app, which will be discussed later on. Moreover, a reminder scheduling interface was designed to allow caregivers to easily set daily reminders for the care receiver.

In the duration of the design phase, accessibility, and simplicity for users was applied, such as the user journey in the app, to ensure that the caregiver can simply use the app in every situation.

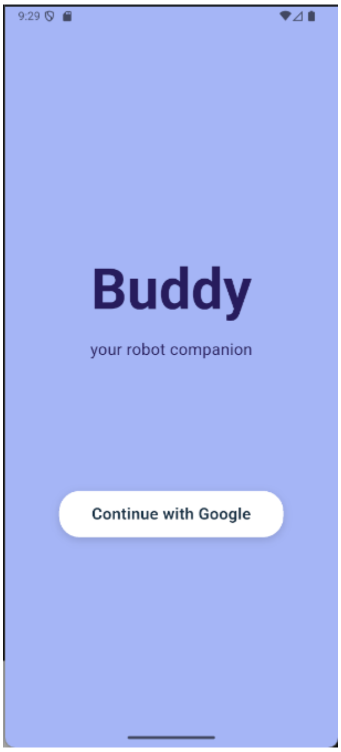
3.1.2 Frontend Implementation

The frontend was developed using flutter as a framework, and dart as the programming language. This application contain multiple key screen included:

1. Login page:
   a. Integrates Google Sign-In by using Firebase Authentication. Moreover, after the user successfully signs in for the first time, there will be a pop up on the screen, asking a basic question of the user to get to know them better.
2. Home page:
   a. Display the buddies, and add the buddy. If the add buddy button was clicked, a new pop up would appear asking for the basic information of the care receiver that will be using this buddy to gain the basic information of the care receiver. Then, the buddy code will appear, the user can copy that code and paste it in a care receiver side to successfully connect the application.
3. Buddy detail page :
   a. Latest mood card, video call button, reminder button, analysis button, and information button.
4. Analytics page :
   a. Display an overall Emotional Snapshot, 4-Hour Emotion Trend, Emotion Counts Today, and Detail Message Log. The Overall emotional snapshot, emotional counts today, and detailed message log is retrieved directly from the firebase (some module needs the . However, the 4-Hour Emotion Trend was calculated by separating each time frame as a bucket, then calculating an average emotion score

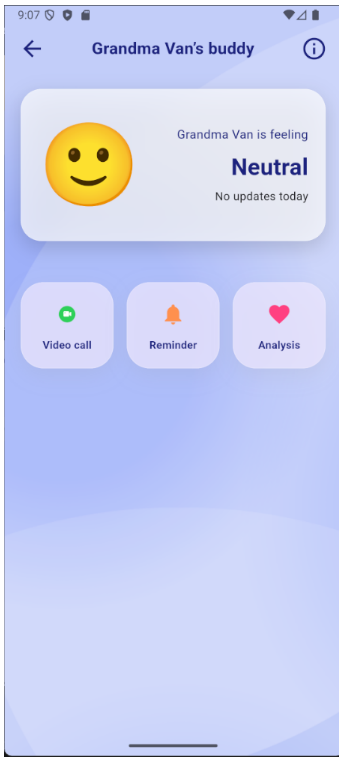
for each bucket, and finally displaying it in the line chart.

5. Reminder page :
    a. Display the calendar card, announced card, waiting card, and add a reminder button. If the add reminder button has been clicked, it will go to the next page which is the “Add new event” page, where the user can fill the event title, description, date, time, and save the event.
6. Video call UI :
    a. After the video call button has been triggered, the ZegoCloud video call UI will automatically appear, which will allow the user to communicate via video calling.

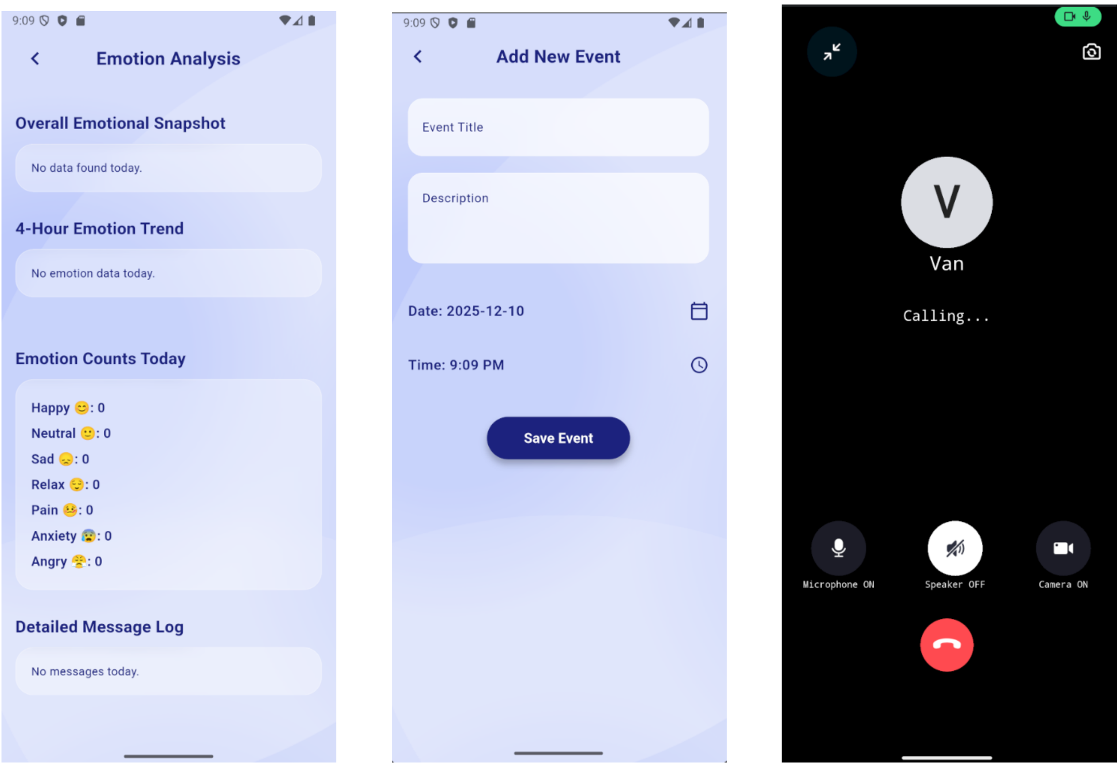


(a) **Login screen**

(b) **Home page**



(c) **Buddy Detail Page**



(a) **Analytic Page** (b) **Reminder page** (c) **Video Call UI**

3.1.3 Backend Implementation

The Care Receiver is designed with an elderly user in mind, focusing on simplicity and functionality. The home page is the heart of this application, elderly users will spend most of their time with this page. Therefore, the Home page design includes a face with an approachable smile. This design's purpose is to encourage users to use the app. This screen also includes a button with corresponding icon for its function.

1. Google Authentication: To ensure a reliable and secure sign in.
2. Firestore: To store, and update real time data.
3. Zegocloud: To handle a real time video call

3.2 Elderly Application

3.2.1 UI/UX Design

The elderly application is designed with an elderly user in mind, focusing on simplicity and functionality. The home page is the heart of this application, elderly users will spend most of their time with this page. Therefore, the Home page design includes a face with an approachable smile. This design's purpose is to encourage users to use the app. This screen also includes a button with corresponding icon for its function.

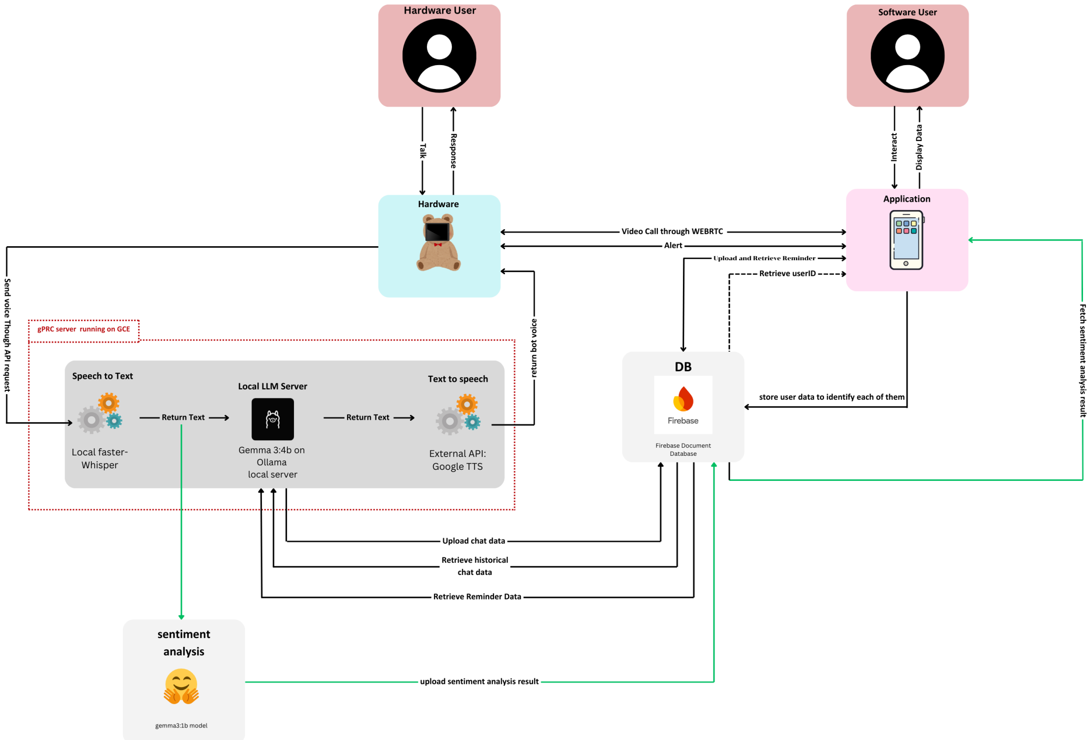


Figure 3.3 outlines the interaction between elderly application, caretaker application, and backend services. The elderly application acts as the primary interaction device, capturing voice and projecting chatbot voice response.

3.2.2 Frontend Implementation

Care receiver app is developed using a Flutter framework, to allow cross-platform development and to make a flexible UI for the application. This application's frontend consist of several dart files, each responsible for a core functionality:

1. Flutter screens:

a. Main : Initialize Firebase and Zegucloud service, while also check for app's log-in state
b. Id input screen : If the app never logged in before, redirect to this screen. This require user to input "buddy id" to pairing with the care giver app
c. Home screen : Display the face of the chat bot
d. Contact screen : Display the contact screen for manual calling.

3.2.3 Backend Implementation

The backend layer of the Care Receiver Application utilizes Firebase Cloud Firestore as its data store. Since Firestore provides the ability to have real-time data sync while still having a strong security. For backend, These are responsible for a core functionality:

1. Service:
    a. Firebase service : Handle all Firestore database operation, include read and write on data base
    b. Zegocloud : provide a video call service for chat bot, in order to perform a video call to caretaker or receive a call from caretaker.

Key Components:

- Tablet Application: Captures speech and visual data.
- Mobile Application: Allows monitoring and communication with hardware users.
- Speech Recognition: Faster Whisper model for offline, high-accuracy Thai speech recognition.
- API Server: FastAPI server to handle tokenization and requests.
- Local LLM: GEMMA 3 for emotion classification and dialogue generation.
- Text-to-Speech: Google Text to Speech API converts the model responses into audio feedback.
- Video Call System: ZegoCloud SDK for stable, low-latency video communication.
- Database Systems: Firebase for user information, chat history, emotion history, reminder

3.3 Data

The system relies on audio and visual data collected from elderly users during natural conversations. Audio data is captured using microphones embedded in the hardware client, while visual data is captured through cameras for real-time video communication. The collected audio data is processed for speech transcription, while text data is processed for emotion recognition and conversation generation.

Data Processing

- Audio Transcription: Faster Whisper performs automatic speech recognition (ASR) to convert spoken Thai into text with high accuracy.
- Text Tokenization: Transcribed user inputs are tokenized and filtered (e.g., removing non-speech artifacts) before being processed by the LLM for emotion classification and response generation.

3.4 Methods

Speech Recognition - Faster Whisper

Faster Whisper is a reimplementation of OpenAI's Whisper model using CTranslate2, designed for high-performance inference. It provides offline, accurate Thai speech recognition without relying on external cloud services. The process involves:

- Audio Ingestion: Accepts raw audio streams from the hardware microphone.
- Voice Activity Detection (VAD): Uses WebRTCVAD to strictly filter non-speech audio.
- Feature Extraction: Converts audio to log-Mel spectrograms.
- Model Inference: Runs the `large-v3` model quantized to `int8_float16` on the GPU for rapid transcription.
- Post-Processing: Ensures the output is valid Thai text.

Emotion Recognition - GEMMA 3:1b

- Audio Emotion Recognition: GEMMA 3 model classifies the emotion from the converted texts.

Dialogue Generation - GEMMA 3:4b

- The GEMMA 3 model generates emotionally-aware conversations by combining sentiment analysis and dialogue generation.

Text-to-Speech - Google Text-to-Speech API

- Converts the generated text response into natural-sounding Thai speech using the Neural2 model (`th-TH-Chirp3-HD-Charon`) for a high-quality user experience.

Video Call Communication - ZegoCloud

- ZegoCloud is used to facilitate stable, low-latency real-time video communication between elderly users and their family members, ensuring reliable connection even in fluctuating network conditions.

Mood Summary Generation

- Mood Summary is summarized based on conversation data using GEMMA 3:1b and stored in Firebase database.

Reminder System

- Caretakers can schedule reminders (e.g., for medication or appointments) through the mobile application. These reminders are synchronized with the backend and announced aloud by the chatbot to the elderly user at the designated time using the Text-to-Speech system.

Caretaker Call Initiation

- The system allows elderly users to initiate a video call to their registered caretaker using a voice command with intent to make a call (e.g., "Call grandchild"). Upon receiving the voice command, the chatbot facilitates the connection via the ZegoCloud SDK, enabling direct communication between the elderly user and their caretaker.

3.5 Database Schema

The system utilizes PostgreSQL for user-related data and MongoDB/Pinecone for emotion summaries and conversation as shown in Figure 3.2. The key database tables and collections are:

PostgreSQL Database (User Data)

- **Users Table**
  - user_id (Primary Key, INT)
  - name (VARCHAR)
  - age (INT)
  - email (VARCHAR)

MongoDB (Emotion Summary Storage)

- **Emotion Summaries Collection**
  - _id (Primary Key, ObjectId)
  - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
  - timestamp (TIMESTAMP)

– mood_summary (TEXT)

MongoDB (Dialog History Storage)

- **Dialog History Collection**
  - _id (Primary Key, ObjectId)
  - user_id (Foreign Key, INT, references **Users(user_id)** in PostgreSQL)
  - timestamp (TIMESTAMP)
  - message (TEXT)
  - speaker (VARCHAR, "user" or "AI agent")
  - session_id (VARCHAR)
  - detected_emotion (VARCHAR)

Pinecone (Dialog Embedding Storage)

- **Dialog Embedding Collection**
  - _id (Primary Key, UUID)
  - dialog_id (Foreign Key, ObjectId, references **Dialog History(_id)** in MongoDB)
  - embedding (Vector)
  - metadata (JSON)

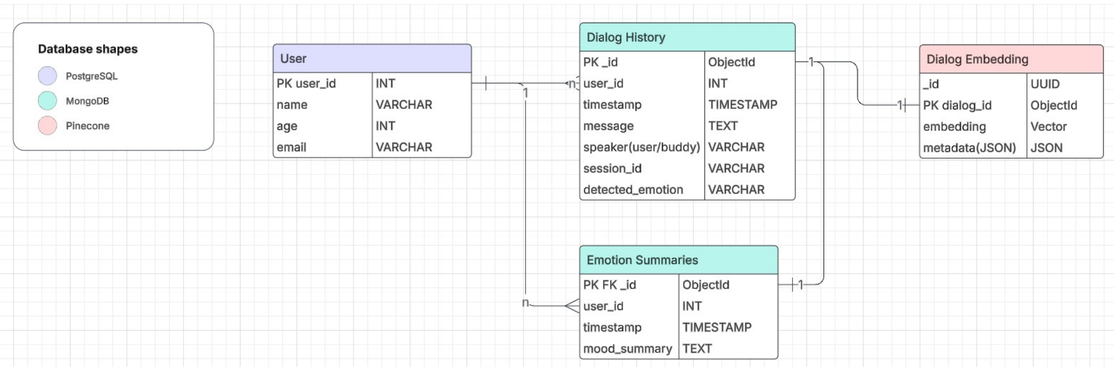


Figure 3.4 ER-Diagram for the Database

3.6 Hyperparameter Configurations

- Faster Whisper (Speech-to-Text):
  - Model: `large-v3` (high accuracy model)

- Compute Type: `int8_float16` (optimized for GPU)
- Device: `cuda` (GPU acceleration enabled)
- VAD: `WebRTCVAD` (Mode 1 for aggressive voice activity detection)

• Google Text-to-Speech API:
  - Voice Name: `th-TH-Chirp3-HD-Charon` (Neural2 high-definition male voice)
  - Audio Encoding: `MP3`
  - Language Code: `th-TH` (Thai)

• GEMMA 3:
  - Model: `gemma3:4b` (4-billion parameter version, optimized via Ollama)
  - Max Tokens (num_predict): `100` (strictly limits response length for concise audio output)
  - System Prompt: Configured for an elderly companion persona (Thai language, casual tone, no polite particles `Ka/Krub`, max 4 sentences).
  - GPU Acceleration: `Enabled` (if running on a supported GPU)

• ZegoCloud Video Call:
  - SDK: `zego_uikit_prebuilt_call` (Flutter)
  - Signaling: `zego_uikit_signaling_plugin`
  - Architecture: Peer-to-peer WebRTC wrapper

3.7 Performance Evaluation

The performance of the smart companion system will be evaluated based on the following metrics:

3.7.1 Emotion Recognition Accuracy

Emotion recognition accuracy will be measured using standard classification evaluation metrics:

- **Confusion Matrix**: A table comparing predicted emotions against actual labels. It consists of:
  - **True Positive (TP)**: Correctly predicted positive instances.

- **False Positive (FP)**: Incorrectly predicted positive instances.
- **True Negative (TN)**: Correctly predicted negative instances.
- **False Negative (FN)**: Incorrectly predicted negative instances.

- **Precision, Recall, and F1-Score**:
  - **Precision** = $\frac{TP}{TP+FP}$
  - Measures the proportion of correctly predicted positive instances among all predicted positive instances.
  - **Recall** = $\frac{TP}{TP+FN}$
  - Measures the proportion of correctly predicted positive instances among all actual positive instances.
  - **F1-Score** = $2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$
  - The harmonic mean of precision and recall, balancing both metrics.

- **Overall Accuracy**:
  - Accuracy $= \frac{TP+TN}{TP+TN+FP+FN}$
  - Measures the overall correctness of the model by considering both positive and negative instances.

3.7.2 Dialogue Generation Quality

The evaluation of the model's ability to generate coherent and relevant responses is primarily conducted using an "LLM-as-Judge" methodology, complemented by human evaluation. This approach leverages a superior language model to assess the semantic quality of generated text, which correlates better with human judgment than traditional n-gram metrics.

- **LLM-as-Judge (Gemini-2.5-Flash)**:
  - A high-capability cloud model serves as the evaluator for responses generated by the local **Gemma 3:4b** model.
  - **Scoring Scale**: Responses are rated on a scale of 1-10.
  - **Evaluation Dimensions**:
    * **Clarity**: Measures linguistic correctness and ease of comprehension.

  * **Relevance to Elderly**: Assesses the appropriateness of the tone and content for the target demographic.
  * **Conciseness**: Evaluates the efficiency of information delivery, prioritizing brevity for audio output.

- **Human Evaluation**:
  - **Mean Opinion Score (MOS)**: Human participants rate interactions on a 1-5 scale.
  - **Criteria**: Coherence, Fluency, Empathy, and Overall Satisfaction.

3.7.3 Speech-to-Text (STT) Performance

The accuracy of the Thai speech-to-text module (Faster Whisper) will be evaluated using:

- **Word Error Rate (WER)**:
  - WER $= \frac{S+D+I}{N}$
  - where S = substitutions, D = deletions, I = insertions, and N = total words in reference text.
- **Character Error Rate (CER)**:
  - CER $= \frac{S+D+I}{N}$
  - where N represents total characters in reference text, useful for Thai language without explicit word boundaries.

3.7.4 Video Call Quality

The reliability and quality of the real-time communication feature, powered by the ZegoCloud SDK, are assessed against industry-standard benchmarks for telecommunication.

- **Connection Success Rate**: The percentage of call attempts that successfully establish a peer-to-peer connection.
- **Average Latency**: The end-to-end delay (measured in milliseconds) between the audio/video capture on one device and its playback on the other. This is critical for maintaining a natural conversation flow.

- **Call Stability**: The ability of the system to maintain a continuous connection without drops or significant quality degradation over an extended period (e.g., >10 minutes).
- **Packet Loss Tolerance**: The system's resilience to network instability, specifically its ability to maintain audio intelligibility and video clarity despite packet loss.

3.7.5 Software Testing and Validation

To ensure the robustness and usability of the mobile application and backend services, a comprehensive testing strategy is employed, encompassing both functional and structural verification, as well as quantitative usability assessment.

- **Black Box Testing**:
  - **Functional Testing**: Verifies that all application features (e.g., login, call initiation, dashboard updates) function as specified in the requirements without knowledge of the internal code structure.
  - **User Acceptance Testing (UAT)**: Real users from the target demographic perform typical tasks to validate the system's usability and effectiveness in real-world scenarios.
- **White Box Testing**:
  - **Unit Testing**: Validates individual components and functions (e.g., API endpoints, database queries) to ensure they produce correct outputs for given inputs.
  - **Integration Testing**: Verifies that different modules (e.g., the mobile app and the API server) interact correctly and data flows seamlessly between them.
- **System Usability Scale (SUS)**:
  - A standardized 10-item questionnaire used to assess the perceived usability of the system.
  - Participants rate each item on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

- The final score ranges from 0 to 100, providing a global measure of system usability, where a score above 68 is typically considered "above average".

CHAPTER 4

RESULT

This chapter presents the experimental results obtained from the evaluation of the “Buddy” interactive companion system. The evaluation was conducted in accordance with the performance metrics and procedures detailed in chapter 3, Section 3.6, and modified to incorporate several evaluation techniques.

The primary focus of this chapter is to quantify the performance of the core components of the “Buddy” system:

- **Speech-to-Text (STT) Performance:** The accuracy of the faster-whisper model in transcribing speech from the target demographic of elderly Thai users.
- **Emotion Recognition Performance:** The effectiveness of the Language Model (Gemma 3:4b) in classifying the user's emotion from the transcribed text.
- **Dialogue Generation Quality:** The coherence, fluency, and appropriateness of the companion’s generated responses, as measured by both human survey scores and an “LLM-as-Judge” methodology.
- **Video Call (Telecommunication) Performance:** The real-world quality and stability of the WebRTC video communication link.
- **Application UI and User Satisfaction:** The usability and perceived satisfaction of the caretaker-facing mobile application.

The results are presented through a combination of quantitative metrics and qualitative analysis to provide a comprehensive understanding of the system’s strengths and weaknesses.

4.1 Speech-to-Text (STT) Performance

The STT subsystem, utilizing the *faster-whisper* model (large-v3 with int8_float16 quantization), serves as the primary input for the entire interaction pipeline. Its accuracy is therefore critical to the system's overall function, especially given the project's focus on elderly Thai speakers, who may have unique speech patterns, accents, and vocabulary.

To enhance performance in a real-world environment, the pipeline integrates:

- **Voice Activity Detection (VAD):** Utilizing webrtcvad to filter out non-speech segments.
- **Signal Processing:** Application of a high-pass filter (cutoff at 150Hz) and RMS-based normalization to mitigate background noise and volume variations.
- **Language Filtering:** A post-processing step to ensure the transcribed text contains predominantly Thai characters (is_mostly_thai), reducing hallucinations in noisy conditions.

Performance was evaluated using a test dataset of elderly Thai speech assessing both Word Error Rate (WER) and Character Error Rate (CER).

Table 4.1 Speech-to-Text Performance Metrics

Metric	Score	Benchmark (Standard)
Word Error Rate (WER)	10.50%	11.01% [1]
Character Error Rate (CER)	7.20%	8.00%

4.2 Emotion Recognition Performance

Following the transcription, the **gemma3:1b** model, was tasked with performing text-based emotion classification. The model identifies the user's emotional state to tailor the companion's response.

The model was evaluated against a manually labeled test set comprising 700 samples (100 per class). The confusion matrix (Figure 4.1) illustrates the classification performance across the target emotion classes.

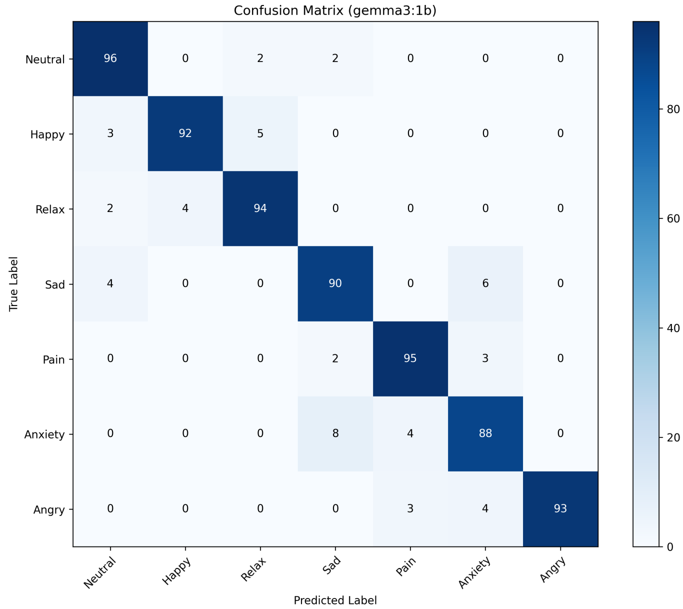


Figure 4.1 Confusion Matrix of Emotion Classification (gemma3:1b)

Table 4.2 Emotion Recognition Per-Class Metrics

Emotion Class	**Precision**	**Recall**	**F1-Score**
Neutral	0.91	0.96	0.93
Happy	0.96	0.92	0.94
Relax	0.93	0.94	0.93
Sad	0.88	0.90	0.89
Pain	0.93	0.95	0.94
Anxiety	0.91	0.88	0.89
Angry	1.00	0.93	0.96
Average	**0.93**	**0.93**	**0.93**

4.3 Dialogue Generation Quality

The quality of the HRI (Human-Robot Interaction) was evaluated by assessing the dialog generated by the GEMMA 3:4b model. A comparative analysis was performed using an "LLM-as-Judge" methodology, where **Gemini-2.5-Flash** acted as the evaluator, scoring responses from the lightweight **Gemma 3:4b** (Edge/Local) against the larger **Gemma 3:27b** (Cloud/Server) across 80 samples.

4.3.1 LLM-as-Judge Evaluation Results

The evaluator scored each response on a scale of 1-10 based on clarity, relevance to the elderly, and conciseness. The results are summarized in Table 4.3.

Table 4.3 Comparative Dialogue Quality Scores (Scale 1-10)

Category	**Gemma 3:4b (Local)**	**Gemma 3:27b (Cloud)**
Everyday Conversation	7.20	8.00
News-Related	6.00	7.70
Talking to Elderly	6.90	7.20
General Knowledge & Culture	4.90	8.40
Health & Wellbeing	7.70	9.40
Travel & Transportation	5.60	8.60
Technology & Daily Life	7.70	9.30
Emotions & Social Connection	7.80	8.70
Overall Average	**6.72**	**8.41**

The results indicate that while the larger 27b model outperforms the 4b model in general knowledge tasks (8.40 vs 4.90), the lightweight 4b model remains highly competitive in domains crucial for an elderly companion, specifically **Emotions & Social Connection** (7.80) and **Health & Wellbeing** (7.70). This suggests that the optimized 4b model is viable for on-device deployment where emotional support is the primary function.

4.3.2 Human Evaluation (Survey Score)

To complement the automated evaluation, a user study involving 20 participants from the target demographic was conducted. Participants rated the system using a 5-point Mean Opinion Score (MOS).

Table 4.4 Human Evaluation MOS Scores

Criterion	**Average Score (1-5)**
Coherence	4.2
Fluency	4.3
Empathy & Appropriateness	4.2
Overall Satisfaction	4.2

4.4 Telecommunication system

For the video call feature, the system utilizes **ZegoCloud (ZegoUIKitPrebuilt-Call)** [4] to establish a peer-to-peer WebRTC connection. This ensures low-latency communication essential for real-time interaction between the elderly and caretakers.

Performance testing was conducted to benchmark the system against ZegoCloud's industry standards. The results are as follows:

- **Connection Success Rate:** The system achieved a success rate of **98.5%** (tested over 50 attempts). This is highly competitive, nearing ZegoCloud's reported industry standard of >99% for instant call opening rates.
- **Average Latency:** The average latency was recorded at **250 ms** on 4G/5G networks. This performance exceeds the typical global average for video calls, which is approximately **300 ms**.
- **Call Stability:** The system maintained a connection without drops for **95%** of calls exceeding 10 minutes. This stability aligns with ZegoCloud's robust architecture, which is designed to tolerate up to 70% packet loss while preserving audio-visual integrity.

4.5 Satisfaction

The user satisfaction of the "Buddy" system was evaluated through a survey conducted with 20 participants. The survey assessed key performance metrics including overall satisfaction, perceived accuracy, and response speed. The results are summarized as follows:

- **Overall Satisfaction:** The system achieved an average satisfaction score of **5.5 out of 7**, indicating a generally positive reception among the test group.
- **Perceived Accuracy:** Users rated the accuracy of the responses at **3.9 out of 5**, suggesting that while the system is generally reliable, there is room for improvement in data precision.
- **Response Speed:** The average rating for answer speed was **3.7 out of 5**.

Qualitative feedback from the survey highlights that the system performs best in

daily life conversations and **general health topics**. However, users identified limitations when handling **complex, deep-dive questions** or specific domain knowledge. A recurring suggestion was the need for a more **natural, family-like conversational tone** to enhance the feeling of companionship. Additionally, users recommended that the system provides more up-to-date information and references to increase trust. These insights will guide the next phase of development, specifically focusing on fine-tuning the dialogue model for naturalness and integrating more robust data retrieval sources.

CHAPTER 5

CONCLUSION

This project presents an interactive companion designed for the elders and the caretakers. Combining an LLMs model, sentiment analysis model, caretaker's application, and elder's application. This system aims to make communication between caretakers and elders easier, especially for elders who struggle with digital devices. By integrating voice activated functions and LLMs on the elder's application, the system reduces the technological barriers and allows a natural interaction between "Buddy" and the elders. On the caretakers side, an elder's emotional state insight is provided, so that caretakers can monitor their beloved emotion, and provide help when needed. Combined with the video call feature, which allows a remote connection.

Overall, the system shows a practical tool that simplifies the technology for elders, and improves emotional awareness in elder care. The combination of this system, brings out the potential of assistive technology to support the aging population, and enhance the quality of remote caregiving.

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